Used Car Price Prediction

Toader Ana-Maria

1 Project overview

The goal of this project is to build a prediction model that estimates the price of used cars based on their specific characteristics. The motivation behind this is to support both buyers and sellers in making informed decisions by leveraging real market data available online.

2 Data acquisition

The dataset was collected from autovit.ro[1] A custom web scraper was implemented to extract vehicle information from HTML pages. The initial raw dataset comprised of 2000 entries with 15 features: make, model, year, body_type, mileage, door_count, nr_seats, color, fuel_type, engine_capacity, engine_power, gearbox, transmission, pollution_standard, and price (target feature). Data collection prioritized completeness, keeping only listings with values for all features, with no duplicate entries found in the dataset.

3 Data filtering

The filtering decisions made were:

- keep only cars manufactured after 2005
- remove cars with mileage above 400,000 km or under 1,000 km (not meaningfully categorized as used cars)
- remove luxury cars exceeding 100,000 km

The outliers visible in the boxplots from Figure 1 are not data errors, but real data representing different market segments. These have been kept, with the goal of contributing to the model's generalization abilities.

After data filtering, the dataset is comprised of 1893 entries, with 13 features: make, model, year, body_type, mileage, color, fuel_type, engine_capacity, engine_power, gearbox, transmission, pollution_standard, and price (target feature).

4 Exploratory data analysis

A comprehensive collection of plots regarding univariate analysis of categorical and numerical features can be found both in the notebooks, as well as in the powerpoint presentation.

[1] www.autovit.ro/autoturisme/second

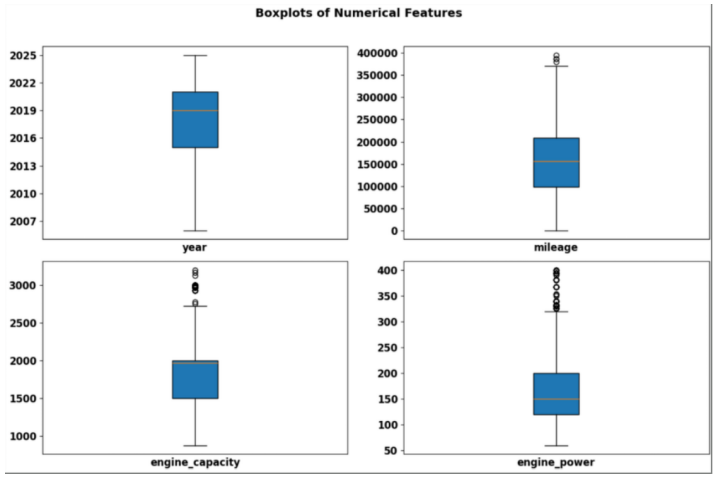

Figure 1: Boxplots of numerical features

Some key observations:

- the price distribution is right-skewed, with low/mid range budget cars vastly outnumbering luxury ones
- feature model has an extreme class imbalance, one-hot encoding would create extremely sparse, imbalanced features
- the other categorical features have a reasonable class imbalance

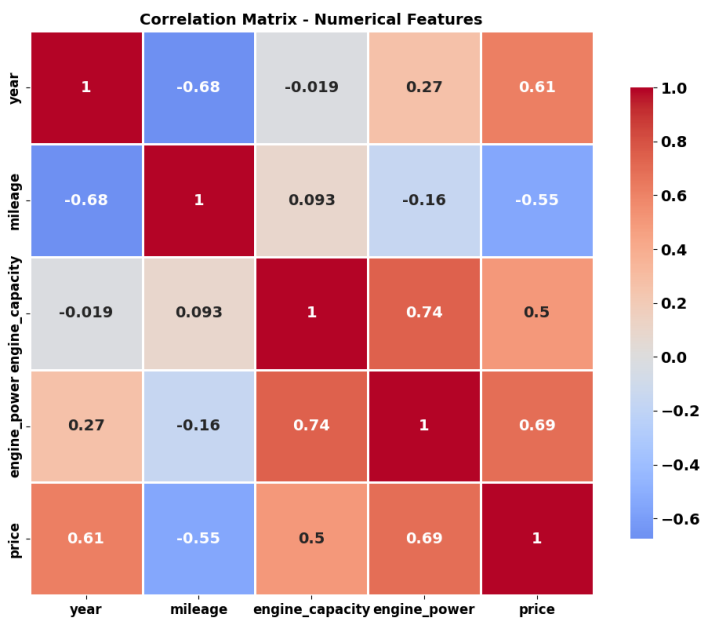

Figure 2: Correlation matrix - numerical features

Figure 2 shows the correlation matrix for numerical features. This shows that engine_capacity and engine_power have a high correlation coefficient, which indicates that using both might be partially redundant, and could cause instability, especially for Linear Regression models.

Tree-based models can handle multicollinearity. Feature engine_power has the highest positive correlation with price, indicating it might be a good predictor.

5 Predictions

The chosen models were Random Forest and XGBoost, because they do not require feature scaling or any complicated transformation, handling the existing dataset with the least amount of preprocessing, and they can handle non-linear relationships.

For the data preprocessing pipeline, the numerical features were passed through unchanged, whereas the categorical features were encoded into integer values using OrdinalEncoder[2].

Hyperparameter tuning was performed using 5-fold cross-validation using GridSearchCV[3].

To ensure a robust model evaluation, the dataset was split into a train-test split with an 80%-20% ratio. Instead of a random split, which could risk creating mismatched distributions, a stratified split was used, by dividing the data into five equal-sized price bins, split proportionately. The goal of this is to obtain balanced distributions, help the model learn from a representative dataset of all price ranges, and ensure evaluation reflects real-world generalization capabilities, particularly across different price segments.

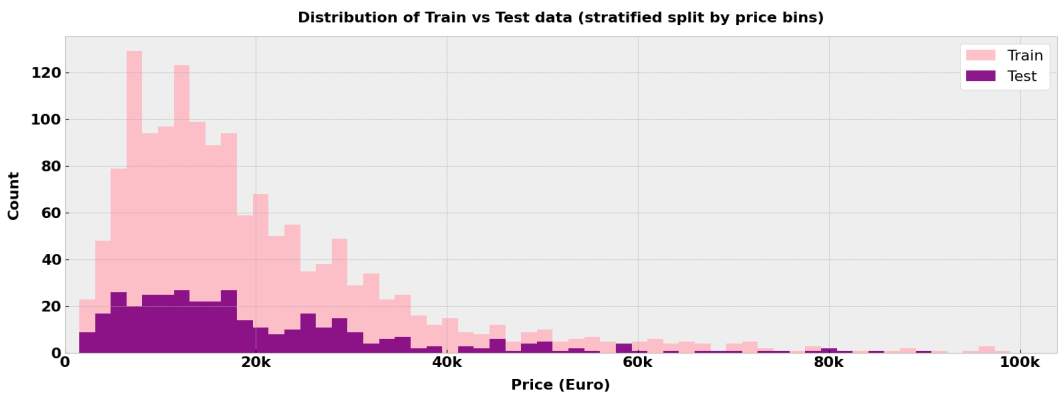

Figure 3: Distributions of train and test data

6 Results

Table 2 shows the results for the best models after the hyperparameter tuning. XGBoost showed superior performance, with a higher R^2 score, and significantly lower values for MAE and RMSE compared to Random Forest.

[2]https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OrdinalEncoder
[3]https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV

Model	R^2	MAE	RMSE
Random Forest	0.837	3697	6474
XGBoost	0.895	3036	5194

Table 1: Results

6.1 Feature importance

After finding the model with its respective hyperparameters that obtained the best scores after the training step, an ablation study was performed to try to determine the importance of features. This procedure is used when calculating "Leave One Feature Out (LOFO) Importance"[4], although in this case it was only used to measure how performance changes.

The values shown in Table 2 are scores obtained after removing one feature at a time from the dataset, and then training and evaluating the XGBoost model on the dataset without that feature. The lowest R^2 score is obtained when the **year** feature is removed, suggesting this is of a high importance. When removing feature **engine_power**, the R^2 score exceeds the value for the base model (trained on the dataset with all the features). This feature is highly collinear with **engine_capacity**, and might introduce noise, becoming redundant.

Feature removed	R^2	MAE	RMSE
make	0.876	3265	5647
model	0.867	3362	5845
year	**0.845**	**3804**	**6315**
body_type	0.886	3168	5414
mileage	0.881	3305	5543
color	0.883	3107	5474
fuel_type	0.892	3068	5271
engine_capacity	0.889	3271	5343
engine_power	**0.909**	**2950**	**4841**
gearbox	0.885	3201	5448
transmission	0.883	3184	5478
pollution_standard	0.886	3216	5413

Table 2: Ablation study: results for the XGBoost model after removing one feature at a time

7 Error analysis

Figure 4 shows a scatter plot for a visual analysis of the model's (XGBoost) predicted price values in relation to the target values from the test dataset. For cars in the lower price range (below 40k euros), it can be seen that the points are more tightly clustered around the diagonal, suggesting lower absolute errors. The error increases significantly for cars above 40k Euro. Looking at the points in the last quantile (above 80k Euro), they tend to fall below the line,

[4] https://christophm.github.io/interpretable-ml-book/lofo.html#fn1

which suggests that the model has a tendency to undervalue expensive cars. This is the most underrepresented price range in the dataset.

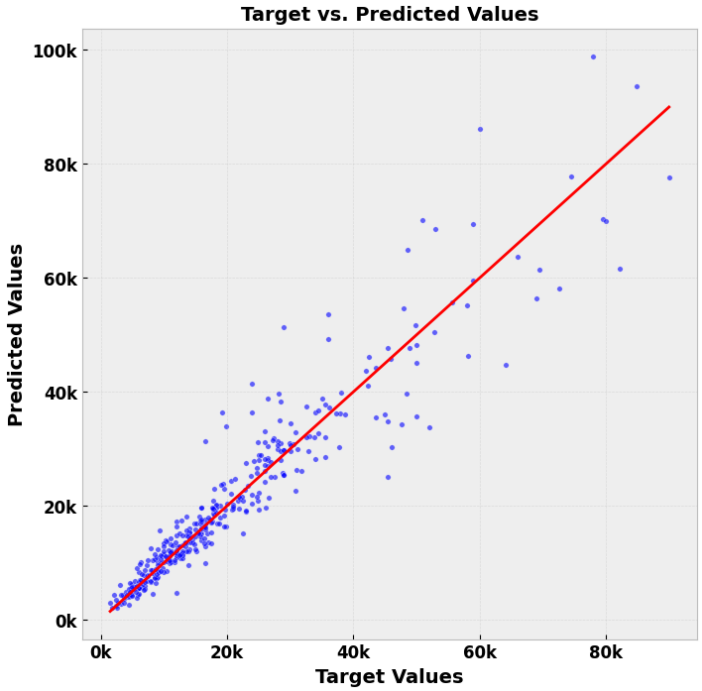

Figure 4: Target price vs. predicted price scatter plot

In conclusion, the model shows a decent performance for mass market cars, and might benefit from additional data or features for premium/luxury cars.